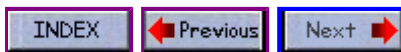


C Programming



KEYBOARD INPUT

There is a function in C which allows the programmer to accept input from a keyboard. The following program illustrates the use of this function,

```
#include <stdio.h>

main()      /* program which introduces keyboard input */
{
    int  number;

    printf("Type in a number \n");
    scanf("%d", &number);
    printf("The number you typed was %d\n", number);
}
```

Sample Program Output

```
Type in a number
23
The number you typed was 23
```

An integer called *number* is defined. A prompt to enter in a number is then printed using the statement

```
printf("Type in a number \n:");
```

The *scanf* routine, which accepts the response, has two arguments. The first ("%d") specifies what type of data type is expected (ie char, int, or float). [List of formatters for scanf\(\) found here.](#)

The second argument (&number) specifies the variable into which the typed response will be placed. In this case the response will be placed into the memory location associated with the variable *number*.

This explains the special significance of the & character (which means the address of).

Sample program illustrating use of scanf() to read integers, characters and floats

```
#include <stdio.h>

main()
{
    int sum;
    char letter;
    float money;

    printf("Please enter an integer value ");
```

```
scanf("%d", &sum );

printf("Please enter a character ");
/* the leading space before the %c ignores space characters in th
scanf(" %c", &letter );

printf("Please enter a float variable ");
scanf("%f", &money );

printf("\nThe variables you entered were\n");
printf("value of sum = %d\n", sum );
printf("value of letter = %c\n", letter );
printf("value of money = %f\n", money );
}
```

Sample Program Output

```
Please enter an integer value
34
Please enter a character
W
Please enter a float variable
32.3
The variables you entered were
value of sum = 34
value of letter = W
value of money = 32.300000
```

This program illustrates several important points.

- the c language provides no error checking for user input. The user is expected to enter the correct data type. For instance, if a user entered a character when an integer value was expected, the program may enter an infinite loop or abort abnormally.
- its up to the programmer to [validate](#) data for correct type and range of values.

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